Amirhossein Bayat

Ph.D. Candidate in Mechanical Engineering
University of Southern Denmark

+45 52 73 96 87 | abay@sdu.dk | bayat-amirhossein.github.io | linkedin.com/in/amir-hossein-bayat-167b21141
| Google Scholar | bayatcw@gmail.com

Profile

Ph.D. candidate in Mechanical Engineering at the University of Southern Denmark, working on topology optimisation for high-heat-flux cooling and turbulent fluid-flow problems. My research combines gradient-based optimisation, finite-element modelling, adjoint sensitivity analysis, conjugate heat transfer, and turbulent-flow simulation to develop efficient computational design methods for advanced thermal-fluid systems.

Education

University of Southern Denmark — *Odense, Denmark*
Ph.D. in Mechanical Engineering — *2024–Present*

- Supervisor: Joe Alexandersen.
- Co-Supervisor: Hao Li.
- Thesis topic: Topology optimisation for high-heat-flux cooling.

Sharif University of Technology — *Tehran, Iran*
M.Sc. in Mechanical Engineering — *2021–2023*

- Supervisor: Ali Hakkaki Fard.
- Thesis title: Solar PV/T collectors enhanced with PEM electrolytic cells.
- CGPA: 18.25/20 (4/4).

Shahid Beheshti University — *Tehran, Iran*
B.Sc. in Mechanical Engineering — *2015–2019*

- CGPA: 17.62/20 (3.8/4).

Selected Research Experience

Topology optimisation for high-heat-flux cooling — *University of Southern Denmark*
Ph.D. research project — *2024–Present*

- Developing computational methods for thermal-fluid topology optimisation with emphasis on cooling performance and manufacturable engineering designs.
- Working with finite-element formulations, adjoint-based sensitivities, PDE filtering, and MMA-based optimisation workflows.
- Extending optimisation methods towards turbulent and conjugate-heat-transfer problems relevant to advanced cooling applications.

Solar PV/T collectors enhanced with PEM electrolytic cells — *Sharif University of Technology*
M.Sc. thesis project — *2021–2023*

- Modelled coupled thermal-energy systems and investigated strategies for improving PV/T collector performance.
- Combined numerical modelling, thermodynamic analysis, and engineering design considerations for renewable-energy applications.

Selected Projects

ASME K-16 Heat Transfer in Electronic Equipment: 2025–2026 Cold Plate Competition — *University of Southern Denmark*
Heat-sink / cold-plate design project — *2025–2026*

- Designed and analysed a compact heat-sink/cold-plate concept for high-heat-flux electronic cooling.
- Combined thermal-fluid simulation and design optimisation ideas to improve heat removal and flow distribution.

Transient temperature modelling of a photovoltaic-thermal cell — *Tehran, Iran*
Sharif University of Technology — *Feb. 2023–Mar. 2023*

- Investigated the outlet-water temperature of a PV/T collector during operating hours.
- Studied the use of phase-change materials and nanofluids to improve simulated cell performance.
- Wrote UDF codes to define time-dependent solar radiation, wind speed, and ambient temperature.

Heat-flux estimation using inverse heat-transfer methods — *Tehran, Iran*
Sharif University of Technology — *Feb. 2023–Mar. 2023*

- Developed Fortran code for inverse heat-transfer calculations and noise-error analysis.
- Created an ANSYS–MATLAB coupled simulation workflow to solve the inverse problem.

Publications and Manuscripts

Density-based topology optimization for Navier–Stokes flow with free-slip boundary conditions — *Structural and Multidisciplinary Optimization, 69(1), Article 23*
Peer-reviewed journal article — *2026*

- Amirhossein Bayat, Hao Li, and Joe Alexandersen.
- DOI: 10.1007/s00158-025-04212-7.

Density-based topology optimization for turbulent fluid flow using the standard k–ϵ RANS model with wall-functions imposed through an implicit wall penalty formulation — *arXiv:2601.02202*
Preprint / manuscript — *2026*

- Amirhossein Bayat, Hao Li, and Joe Alexandersen.
- Focus: turbulent fluid-flow topology optimisation using RANS modelling and implicit wall-function treatment.

Research Interests

Core areas	Fluid mechanics, turbulent fluid simulation, heat transfer, topology optimisation, conjugate heat transfer, and high-heat-flux cooling.
Methods	Finite element method, adjoint sensitivity analysis, PDE-constrained optimisation, RANS turbulence modelling, wall-function modelling, and MMA-based optimisation.
Emerging interests	Microfluidics, particle separation, acoustofluidics, and thermal-fluid design for fusion-energy applications.

Teaching Experience

Structural Engineering, M.Sc. course — *University of Southern Denmark*
Teaching Assistant — *Spring 2025*

- Supported master's-level teaching activities, exercises, and student learning in structural engineering.

Advanced Finite Element Method, M.Sc. course — *University of Southern Denmark*
Teaching Assistant — *Fall 2025*

- Supported master's-level teaching activities, exercises, and student learning in advanced finite-element analysis.

Heat-transfer experiments for B.Sc. students — *University of Southern Denmark*
Laboratory and Experimental Teaching Support — *2024–2026*

- Responsible for preparing, supporting, and guiding undergraduate heat-transfer laboratory experiments.

Supervision Experience

Jens Ole Jepsen, M.Sc. student — *University of Southern Denmark*
Co-supervisor — *2024–2025*

- Co-supervised the master's thesis: *Topology optimisation and experimental testing of twisted-tape cooling inserts for straight cooling channels*.
- Supported the student in topology optimisation, thermal-fluid modelling, experimental planning, and interpretation of heat-transfer results.
- The student graduated in 2025.

EMİRHAN SAĞDIÇ, B.Sc. student *University of Southern Denmark*
Co-supervisor *2026*

- Supervised a bachelor's project on topology optimisation of cooling-channel insert designs.
- Guided the student on computational modelling, design interpretation, and heat-transfer-focused optimisation ideas.

Conferences and Symposia

World Congress of Structural and Multidisciplinary Optimization (WCSMO) *Kobe, Japan*
Oral presentation *2025*

- Presented research on density-based topology optimisation for fluid-flow problems.

WCCM–ECCOMAS 2026 *Munich, Germany*
Oral presentation *2026*

- Presentation at the 17th World Congress on Computational Mechanics and 10th European Congress on Computational Methods in Applied Sciences and Engineering.

DCAMM Internal Symposium *Denmark*
Poster presentation *2024*

- Presented Ph.D. research progress in applied mathematics and mechanics.

DCAMM Internal Symposium *Denmark*
Oral presentation *2026*

- Presented research progress in topology optimisation and thermal-fluid modelling.

Granted Funds

THOMAS B. THRIGES FOND — WCSMO Congress, Kobe, Japan *Grant reference: 5104-2411*
Travel grant, DKK 20,000 *Nov. 2024–Jul. 2025*

- Awarded support for participation in the Congress of Structural and Multidisciplinary Optimization.
- Funding agency: THOMAS B. THRIGES FOND.

DANfusion — WCCM–ECCOMAS 2026, Munich, Germany *DANfusion, Denmark*
Travel support, DKK 12,000 *2026*

- Awarded funding support for participation in WCCM–ECCOMAS 2026, connected to fusion-energy research.
- DANfusion travel support is intended for students and scientists at Danish universities in relation to fusion-energy research.

Technical Skills

Programming	MATLAB, Python, Fortran, FreeFEM++, Firedrake.
Simulation	COMSOL Multiphysics, ANSYS Fluent, Abaqus, Tecplot, PipeNet.
Numerical methods	FEM, turbulent-flow simulation, topology optimisation, adjoint sensitivities, and PDE filtering.
CAD and tools	SolidWorks, CATIA, Inventor, AutoCAD, LaTeX/Overleaf, Microsoft Office.
Professional	Scientific writing, documentation, teamwork, problem solving, and research presentation.

Languages

English	Professional proficiency, C1.
Persian	Native proficiency.
Danish	Elementary proficiency, A2.

References

Joe Alexandersen University of Southern Denmark; Phone: +45 65 50 74 65; Email: joal@sdu.dk.

Hao Li University of Southern Denmark; Phone: +86 13 61 16 32 763; Email: hli@sdu.dk.